



Research Methods for IT Professionals (REM475S)

ADVANCED DIPLOMA INFORMATION TECHNOLOGY

DR LYNN COLEMAN, MONDAY 3 FEBRUARY 2020

Some questions?

- Why are you here?
- What is research?
- What is a researcher? What do they look like?
- What do you think you will 'learn' in this subject?
- How will you 'learn' the content of this subject?



Scientific or Academic Research

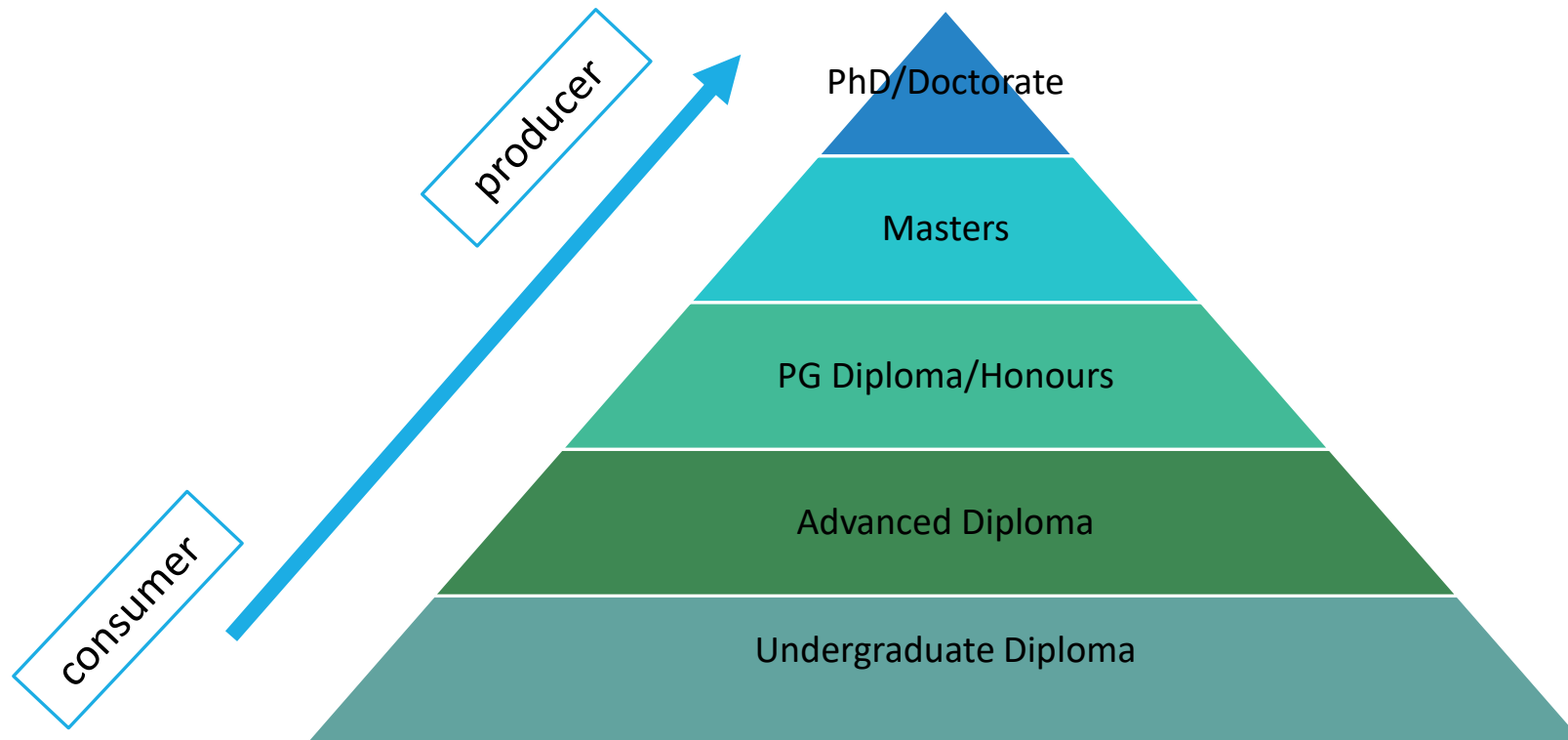
- How we know what we know?
- How civilisation and society has developed?
- How we are able to separate truth from falsehoods?
- How we solve problems?



FACT or OPINION



Consumer or Producer





Research Methods for IT Professionals (REM475S)

NUTS AND BOLTS

WHAT YOU HAVE TO DO

1. Propose, design and conduct a small-scale research investigation, located within the IT professional context using basic questionnaire design
2. Produce and create various communication documents (written, oral, visual) that demonstrate appropriate academic conventions and practices
3. Exemplify appropriate ethical standards/values associated with academic research
4. Describe, illustrate and examine at an introductory level the different methodologies, methods, processes, procedures and techniques associated with legitimate academic/scientific research

WHAT I WILL DO

1. Introduce students to the knowledge, principles, techniques, methods and philosophies associated with scientific/academic research as it applies to the IT disciplinary and professional contexts
2. Guide student through the design, planning, implementation (conducting) and reporting of a small-scale research investigation, using basic questionnaire design and located in an IT industry/professional context
3. Explore and discuss the different academic conventions and practices associated with reporting research findings
4. Provide practice opportunities to produce different academic documents associated with a research undertaking e.g. written reports, oral presentations
5. Introduce students to the principles, values and behaviours of an academic researcher that shown understanding of and adheres to the basic ethical standards and prescriptions

What will classes be like

During our classroom sessions you will be required to become an **active learner**; this means you have to do things; like take notes, respond to questions, discuss with your peers, do written activities, work on a computer, read, work with your peers in groups, speak to your peers and your lecturer, think and concentrate. You will have to complete the **preparatory reading and/or writing tasks before coming to class** – this will ensure you are able to engage with the subject content in a deep and intelligent way, and so gain the most from your very short time with your lecturer.

Demonstrating your learning

In this subject you will be asked to undertake different kinds of tasks and activities so that you can demonstrate that you have learnt and understood the content, topics and skills taught in the classroom.

The assessments in this subject will ask you to demonstrate what you KNOW, show that you can APPLY or USE this knowledge in practice, while also express your BELIEFS and VALUES about scientific research.

It is very important to understand that your understanding, interpretation and application of the various concepts, guidelines, standards and approaches are more important than your ability to memorise facts.

Look carefully at the distribution of marks for each assignment – most of the assignments require you to apply your knowledge and understanding to written research projects or reports.

You will receive assessment/learning feedback after each assignment you complete. Feedback will be presented either individually or more generally to the whole class. These sessions are crucial for helping you establish what you are doing well and the areas that need more improvement.

Academic Writing

Academic writing is seen as central to your learning in this subject. You have to take reading and writing activities very seriously. You will also be provided with multiple opportunities to improve your academic writing, especially for the assessment requirements in this subject.



Assessment

Semester	Types of Assessment	Nature of Assessment	Assessment Weight
ONE	1. Close-book exam	Individual	10
	2. Written research proposal	Individual	20
	3. Reflective piece (1)	Individual	10
TWO	4. Research Report (part 1 & 2)	Individual/Pair	Part 1 = 20 Part 2 = 20
	5. Research Presentation	Individual/Pair	10
	6. Reflective piece (2)	Individual	10

40%

60%

Assessment

Semester	Types of Assessment	Assessment Weight	Proposed Dates
ONE	1. Close-book exam	10	March 2020
	2. Written research proposal	20	May/June 2020
	3. Reflective piece (1)	10	May/June 2020
TWO	4. Research Report (part 1 & 2)	Part 1 = 20 Part 2 = 20	Mid-August 2020 Early-October 2020
	5. Research Presentation	10	Mid-October 2020
	6. Reflective piece (2)	10	Mid-October 2020

Learning Resources

Main Textbooks:

Kumar, Ranji, 2011 – Research Methodology: A step by step guide for beginners, 3rd Edition (available as downloadable PDF). 2014 4th Edition available for purchase

Supplementary Textbooks:

1. **Booth et al, 2016** – The Craft of Research, University of Chicago Press: Chicago, ISBN-13: 9780226239736
2. **Terre Blanche, M and Durrheim, K and Painter, D. (editors) 2006** – Research in Practice 2nd Edition, UCT Press: Cape Town, ISBN: 9781919713694
3. **Blaxter et al, 2006** – How to research 3rd Ed, Open University Press: Maidenhead, ISBN: 033521746X
4. **Maree, 2016** – First steps in research 2nd Ed Chapter, Van Schaik: Pretoria, ISBN: 9780627033698

Next week

1. Read through subject guide

- Bring all your questions, queries to class

2. Read Booth et al reading (Chapter 1: Thinking in Print)

- **Answer the following questions**
 - What is research?
 - How is research described?
 - What are the pillars of scientific research?
 - Why do we 'write research'?
- **Come prepared to answer these questions in class**